Monopolistic Competition

EC 111 Principles of Microeconomics

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Reading

- These slides are not a substitute for reading Mankiw, Chapter 17, Monopolistic Competition.
- Required reading: Mankiw, *Principles of Microeconomics*, 11th edition, Chapter 17, "Monopolistic Competition".

What you must know from this chapter

1. Definition: many firms, differentiated products, free entry.
2. Why each firm faces a downward-sloping demand curve.
3. The short run: MR $=$ MC, the price from demand, profit or loss on the graph.
4. The long run: entry and exit drive profit to zero, $P =$ ATC.
5. Comparison with perfect competition: excess capacity and a markup, $P >$ MC.

Part 1

What Is Monopolistic Competition?

Monopolistic competition

Definition: Monopolistic competition

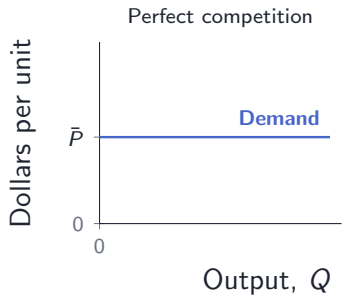
A market with many firms selling products that are similar but not identical, and free entry and exit.

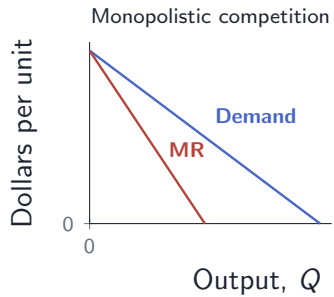
- **Many sellers:** many firms compete for the same group of buyers.
- **Product differentiation:** each firm's product is slightly different from its rivals'.
- **Free entry and exit:** firms can enter or leave the market without restriction.

Examples: restaurants, hair salons, coffee shops, clothing brands.

Each firm faces a downward-sloping demand curve

- A restaurant that raises its price loses some customers to rivals.
- It does not lose them all: some prefer its food.
- So its demand curve slopes down.
- As for a monopoly, $P > \text{MR}$.





Between perfect competition and monopoly

	Perfect competition	Monopolistic competition	Monopoly
Number of firms	Many	Many	One
Product	Identical	Differentiated	No close substitutes
Entry	Free	Free	Blocked
Firm's demand	Horizontal	Slopes down	Slopes down

Finding

Like perfect competition: many firms and free entry. Like monopoly: a downward-sloping demand curve, so $P >$ MR.

Part 1 summary

- Monopolistic competition: many firms, differentiated products, free entry and exit.
- Each firm's product differs a little from its rivals'.
- So each firm faces a downward-sloping demand curve.
- So, as for a monopoly, $P >$ MR.
- Compared with perfect competition: the same many firms and free entry, but differentiated products.

Your turn: Check your understanding

1. Which market is monopolistically competitive?
   A. Wheat farming. B. Hair salons in a city. C. A town's water network.
2. A hair salon raises its price. It loses:
   A. none of its customers. B. some of its customers. C. all of its customers.
3. New salons can open whenever they like.
   A. True B. False

Solution: Check your understanding

1. **B.** Many salons, each slightly different, free entry. Wheat is identical (perfect competition); the water network is a natural monopoly.
2. **B.** Some customers prefer this salon, so not all leave. Demand slopes down.
3. **A.** True: entry is free.

Check: Part 1

A monopolistically competitive industry is characterized by

A many firms, differentiated products, and barriers to entry.

B many firms, differentiated products, and free entry.

C a few firms, identical products, and free entry.

D a few firms, differentiated products, and barriers to entry.

Check: Part 1 – answer

A monopolistically competitive industry is characterized by

A many firms, differentiated products, and barriers to entry.

B many firms, differentiated products, and free entry.

C a few firms, identical products, and free entry.

D a few firms, differentiated products, and barriers to entry.

Finding

The three features of monopolistic competition: many firms, differentiated products and free entry.

Part 2

The Short Run: Act Like a Monopoly

The same three steps as a monopoly

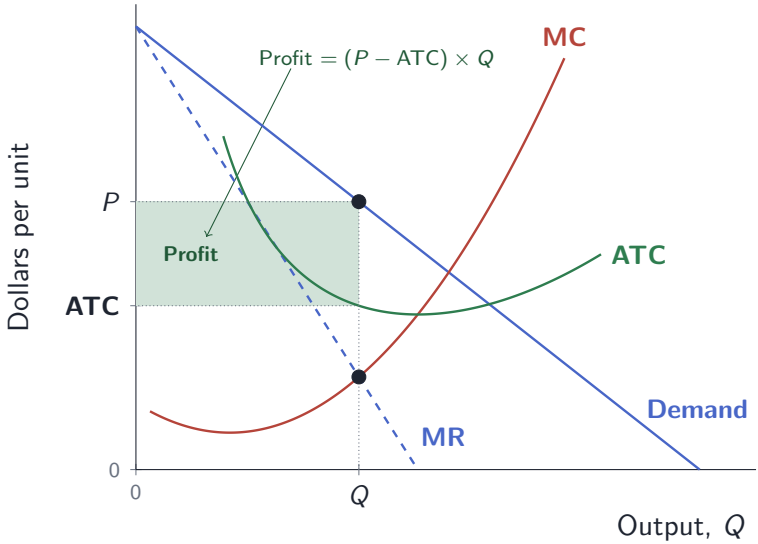
1. **Output:** produce Q where MR $=$ MC.
2. **Price:** go up from Q to the demand curve. That is the price, P.
3. **Profit:** compare P with ATC at Q:

$$\text{Profit} = (P - \text{ATC}) \times Q.$$

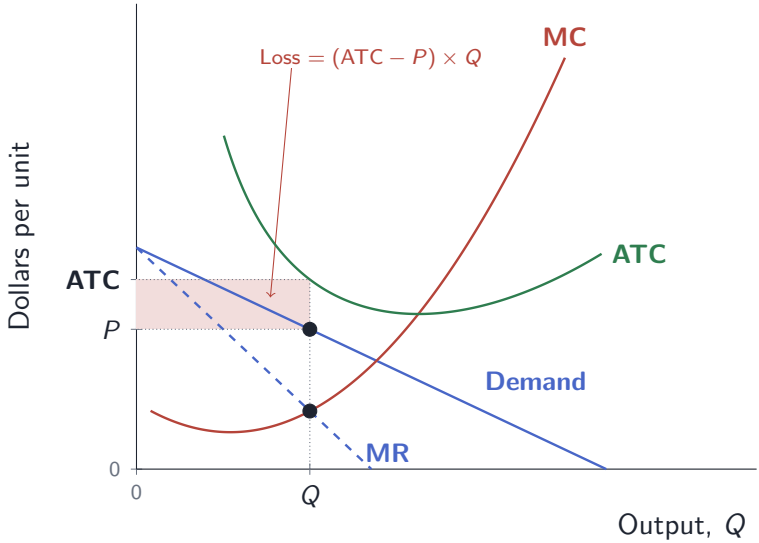
Finding

If $P >$ ATC, the firm makes a profit. If $P <$ ATC, it makes a loss.

Short-run profit

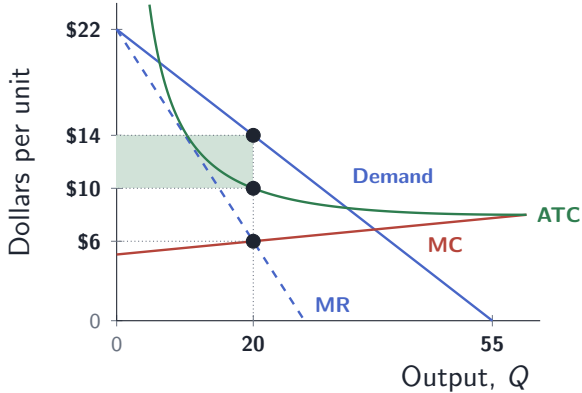


Short-run loss



Example: reading profit from the graph

1. MR = MC at $Q = 20$.
2. Up to demand: $P = \$14$.
3. ATC at 20 units: \$10.
4. Profit $= (14 - 10) \times 20 = \$80$.



Source of the numbers: demand $P = 22 - 0.4Q$; TC $= 90 + 5Q + Q^2/40$, so MC $= 5 + Q/20$ and ATC $= 90/Q + 5 + Q/40$.

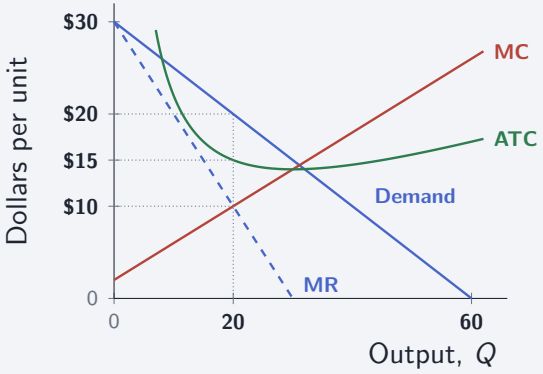
Part 2 summary

- In the short run the firm acts like a monopoly.
- Output: where MR = MC.
- Price: read from the demand curve at that output.
- Profit = $(P - \text{ATC}) \times Q$.
- If $P < \text{ATC}$, the firm makes a loss of $(\text{ATC} - P) \times Q$.

Your turn: Check your understanding

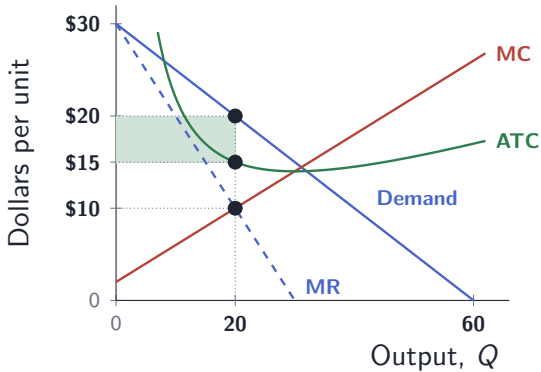
The firm maximizes profit.

1. Its output is:
   A. 10 B. 20 C. 60
2. Its price is:
   A. \$10 B. \$15 C. \$20
3. Its profit is:
   A. \$100 B. \$200 C. \$300



Solution: Check your understanding

1. **B.** MR meets MC at 20 units (both \$10).
2. **C.** Up to demand at 20 units: \$20.
3. **A.** ATC at 20 is \$15. Profit $= (20 - 15) \times 20 = \$100$.



Check: Part 2

In the short run, a firm operating in a monopolistically competitive market

A produces an output level where marginal revenue equals average total cost.

B produces an output where marginal revenue equals marginal cost, and the price is determined by demand.

C must earn zero economic profits.

D maximizes revenues as well as profits.

Check: Part 2 – answer

In the short run, a firm operating in a monopolistically competitive market

A produces an output level where marginal revenue equals average total cost.

B produces an output where marginal revenue equals marginal cost, and the price is determined by demand.

C must earn zero economic profits.

D maximizes revenues as well as profits.

Finding

Steps 1 and 2: output where MR = MC, then the price from the demand curve.

Part 3

The Long Run: Entry and Exit

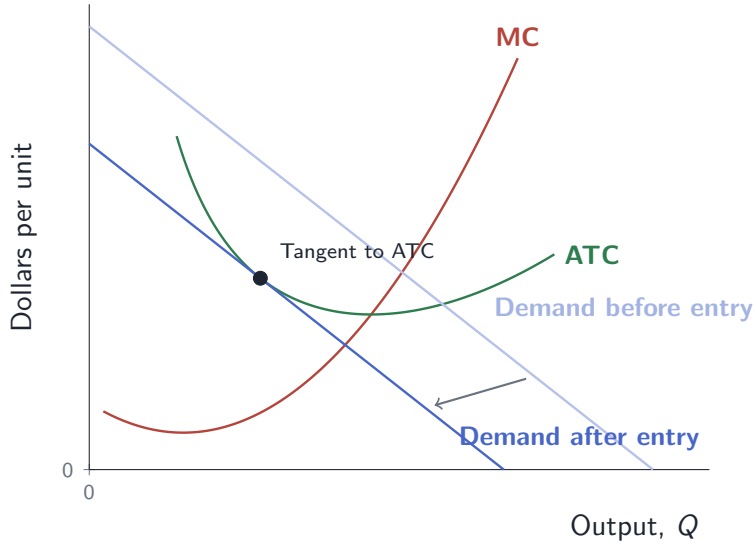
Profit attracts entry; losses cause exit

- **Profit:** new firms enter. Buyers now have more products to choose from.
- Each existing firm loses some customers: its demand curve shifts left.
- Entry continues until profit is zero.
- **Losses:** some firms exit. Their customers move to the firms that remain.
- Each remaining firm's demand curve shifts right, until the loss is zero.

Finding

Entry and exit move each firm's demand curve until profit is zero.

Entry on the graph



Long-run equilibrium: two conditions

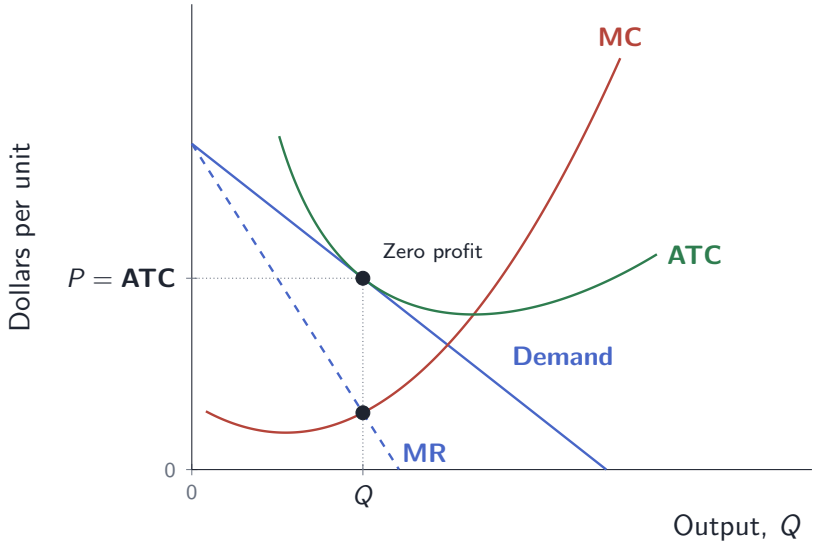
1. **The firm maximizes profit:** MR = MC.
2. **Profit is zero:** P = ATC.

- Profit = $(P - \text{ATC}) \times Q = 0 \times Q = 0$.
- On the graph, the demand curve just touches the ATC curve (it is tangent to ATC).

Finding

In the long run, MR = MC and P = ATC: demand is tangent to ATC and profit is zero.

Long-run equilibrium on the graph



Part 3 summary

- Profit attracts entry: each firm's demand shifts left.
- Losses cause exit: each remaining firm's demand shifts right.
- This stops when profit is zero.
- Long run: MR $=$ MC and $P =$ ATC.
- On the graph, demand is tangent to ATC.
- Compared with perfect competition: long-run profit is zero in both.

Your turn: Check your understanding

Coffee shops in a city are making a profit.

1. New shops open. Each existing shop's demand curve:
   A. shifts left. B. shifts right. C. does not move.
2. In the long run, each shop's profit is:
   A. positive. B. zero. C. negative.
3. In the long run, a shop sells 300 cups a day at \$4 a cup. Its ATC is:
   A. \$3. B. \$4. C. \$5.

Solution: Check your understanding

1. **A.** Buyers spread over more shops, so each shop sells less at every price.
2. **B.** Entry continues until profit is zero.
3. **B.** Zero profit means $P = \text{ATC}$, so ATC $= \$4$.

Check: Part 3

In monopolistically competitive markets, free entry and exit suggests that

A the market structure will eventually be characterized by perfect competition in the long run.

B all firms earn zero economic profits in the long run.

C some firms will be able to earn economic profits in the long run.

D some firms will be forced to incur economic losses in the long run.

Check: Part 3 – answer

In monopolistically competitive markets, free entry and exit suggests that

A the market structure will eventually be characterized by perfect competition in the long run.

B all firms earn zero economic profits in the long run.

C some firms will be able to earn economic profits in the long run.

D some firms will be forced to incur economic losses in the long run.

Finding

Entry removes profit and exit removes losses, so long-run profit is zero.

Part 4

Monopolistic Competition versus Perfect Competition

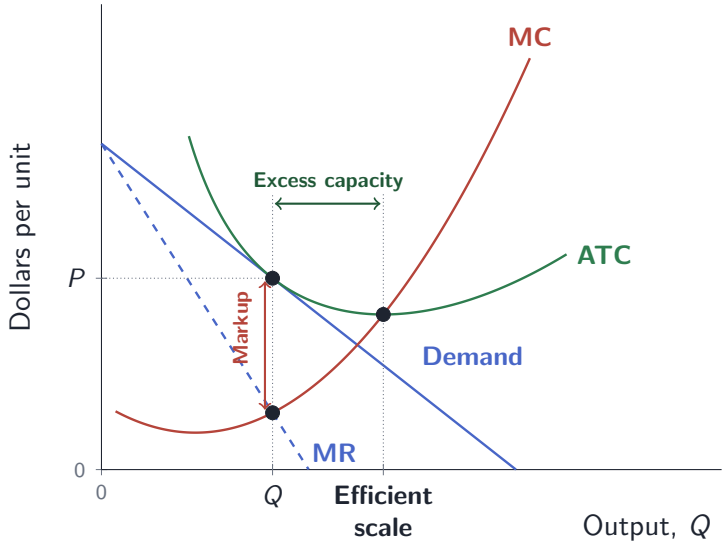
Two differences in the long run

Definition: Efficient scale

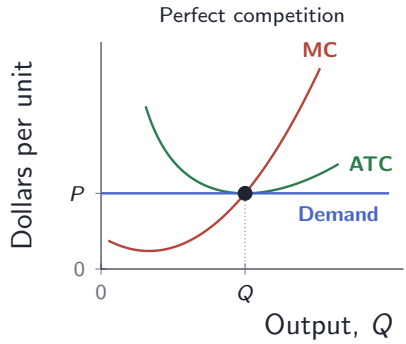
The output at which average total cost is at its minimum.

- A competitive firm produces at the efficient scale, with $P = \text{MC}$.
- **Excess capacity:** a monopolistically competitive firm produces *less* than the efficient scale.
- **Markup:** it charges a price above marginal cost, $P > \text{MC}$. The markup is $P - \text{MC}$.

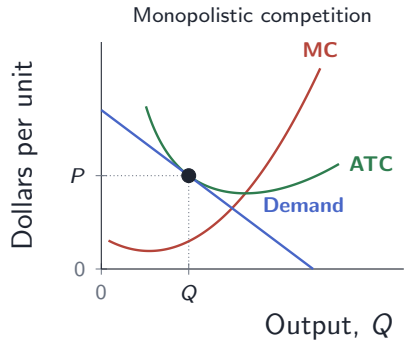
Excess capacity and markup on the graph



The two long runs side by side



$P = \text{MC}$. Output at the efficient scale.



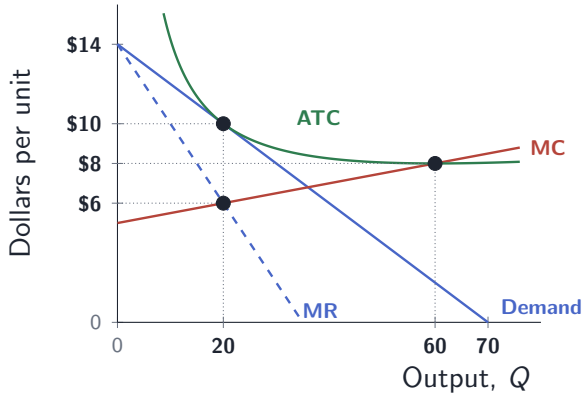
$P > \text{MC}$. Output below the efficient scale.

Finding

Profit is zero in both. Only perfect competition reaches the efficient scale with $P = \text{MC}$.

Example: a firm in long-run equilibrium

1. MR = MC at $Q = 20$; MC = \$6.
2. Demand at 20: $P = \$10 =$ ATC. Profit is zero.
3. Markup $= P - \text{MC} = 10 - 6 = \$4$.
4. Efficient scale: 60 units, ATC = \$8.
5. Excess capacity $= 60 - 20 = 40$ units.



Source of the numbers: demand $P = 14 - 0.2Q$; $\text{TC} = 90 + 5Q + Q^2/40$, so $\text{MC} = 5 + Q/20$ and $\text{ATC} = 90/Q + 5 + Q/40$.

The comparison in a table

Long run	Perfect competition	Monopolistic competition
Profit	Zero	Zero
Price and ATC	$P = \text{ATC}$	$P = \text{ATC}$
Price and MC	$P = \text{MC}$	$P > \text{MC}$: a markup
Output and efficient scale	At the efficient scale	Below it: excess capacity

Finding

Because $P > \text{MC}$, some buyers who value a unit above its cost do not buy it. That is why monopolistic competition is inefficient.

Part 4 summary

- Efficient scale: the output at minimum ATC.
- Excess capacity: the firm produces less than the efficient scale.
- Markup: the firm charges $P > \text{MC}$; the markup is $P - \text{MC}$.
- Long-run profit is zero, as in perfect competition.
- Compared with perfect competition: a competitive firm produces at the efficient scale with $P = \text{MC}$.

Your turn: Check your understanding

A firm in long-run equilibrium sells 20 units at \$15. Its MC at 20 units is \$10. Its ATC is lowest, \$14, at 30 units.

1. The markup is:
   A. \$1 B. \$5 C. \$15
2. The excess capacity is:
   A. 0 units B. 10 units C. 30 units
3. A competitive firm with these costs would, in the long run, charge:
   A. \$10 B. \$14 C. \$15

Solution: Check your understanding

1. **B.** Markup $= P - \text{MC} = 15 - 10 = \$5$.
2. **B.** Efficient scale minus output: $30 - 20 = 10$ units.
3. **B.** In the long run a competitive firm produces at the efficient scale, so $P =$ minimum ATC $= \$14$.

Check: Part 4

In a long-run equilibrium,

A only a perfectly competitive firm operates at its efficient scale.

B only a monopolistically competitive firm operates at its efficient scale.

C neither a competitive firm nor a monopolistically competitive firm charges a markup over marginal cost.

D both a perfectly competitive firm and a monopolistically competitive firm operate at their efficient scale of production.

Check: Part 4 – answer

In a long-run equilibrium,

A **only a perfectly competitive firm operates at its efficient scale.**

B only a monopolistically competitive firm operates at its efficient scale.

C neither a competitive firm nor a monopolistically competitive firm charges a markup over marginal cost.

D both a perfectly competitive firm and a monopolistically competitive firm operate at their efficient scale of production.

Finding

The monopolistically competitive firm has excess capacity: it produces below the efficient scale.

Which rule answers which question

Question	Rule
What defines the market?	Many firms, differentiated products, free entry.
What demand does a firm face?	Downward-sloping, so $P > \text{MR}$.
How much does it produce?	$\text{MR} = \text{MC}$.
What price does it charge?	The price on the demand curve at that output.
Short-run profit?	$(P - \text{ATC}) \times Q$; a loss if $P < \text{ATC}$.
Long-run profit?	Zero: $P = \text{ATC}$, demand tangent to ATC.
How does it compare with competition?	Excess capacity and a markup, $P > \text{MC}$.